

EMBOSS NEEDLE

Pairwise Sequence Alignment (PSA)

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Results for Job ID

emboss_needle-I20260219-005309-0305-13593901-p2m

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Resubmission



Tool Output

Result Files

Submission Details

Tool output

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```
#####
# Program: needle
# Rundate: Thu 19 Feb 2026 00:53:11
# Commandline: needle
#   -auto
#   -stdout
#   -asequence emboss_needle-I20260219-005309-0305-13593901-p2m.asequence
#   -bsequence emboss_needle-I20260219-005309-0305-13593901-p2m.bsequence
#   -datafile EDNAFULL
#   -gapopen 10.0
#   -gapextend 1.0
#   -endweight
#   -endopen 10.0
#   -endextend 1.0
#   -aformat3 pair
#   -snucleotide1
#   -snucleotide2
# Align_format: pair
# Report_file: stdout
#####

#=====
#
# Aligned_sequences: 2
# 1: NM_002124.3
# 2: NM_001265803.1
# Matrix: EDNAFULL
# Gap_penalty: 10.0
# Extend_penalty: 1.0
#
# Length: 1396
# Identity:   574/1396 (41.1%)
# Similarity: 574/1396 (41.1%)
# Gaps:      643/1396 (46.1%)
# Score: 548.0
#
#
#=====

NM_002124.3      1 CCTATAACTTGGGAATGTGGGTGGAGGGGTTCATAGTTCTCCCTGAGTGAG      50
                  ||      ||                      |||
NM_001265803.   1 CC-----GG-----CAGCC-----      9

NM_002124.3     51 ACTTGCCTGCTTCTCTGGCCCTGGTCCTGTCCTGTTCTCCAGCATGG--      98
                   |||||..|||.||  ||  |||.||..|||.||
NM_001265803.   10 ----GCCTGCTCTTCTCGC----GG---GTCCAGACGCGAGCGCGGGG      47

NM_002124.3     99 -----TGTGTCTGAAG-CTC-----CCTGGAGGCTCCTGCATG---    130
                   ||.|||  | |||  ||.||.||| .|.|||
```


LALIGN

Pairwise Sequence Alignment (PSA)

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Results for Job ID

lalign-I20260219-020217-0691-35370058-p1m

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Resubmission



Tool Output	Result Files	Submission Details
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Tool output

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```
# /fasta/bin/lalign36 -m 9i lalign-I20260219-020217-0691-35370058-p1m.asequence lalign-I20260219-020217-0691-35370058-p1m.bsequence -n -s BL50 -f -12 -g 0 -E 0.01 -m 0 -m "F11 lalign-I20260219-020217-0691-35370058-p1m.output.lav"
LALIGN finds non-overlapping local alignments
version 36.3.8h May, 2020
Please cite:
X. Huang and W. Miller (1991) Adv. Appl. Math. 12:373-381

Query: lalign-I20260219-020217-0691-35370058-p1m.asequence
1>>>Rosalind_0472 - 883 nt
Library: lalign-I20260219-020217-0691-35370058-p1m.bsequence
      868 residues in      1 sequences

Statistics: (shuffled [500]) MLE statistics: Lambda= 0.0367; K=5.569e+27
statistics sampled from 1 (1) to 500 sequences
Threshold: E() < 0.01 score: 2234
Algorithm: Smith-Waterman (SSE2, Michael Farrar 2006) (7.2 Nov 2010)
Parameters: BL50 matrix (13:-5), open/ext: -12/0
Scan time: 1.790

The best non-identical alignments are:      1s-w bits E(1) %_id %_sim alen
Rosalind_9829              ( 868) [f] 2395 34.7 2.7e-05 0.461 0.521 1040
+-
      2238 26.4 0.0087 0.438 0.511 1024

>>>Rosalind_0472, 883 nt vs lalign-I20260219-020217-0691-35370058-p1m.bsequence library

>>Rosalind_9829              (868 nt)
Waterman-Eggert score: 2395; 34.7 bits; E(1) < 2.7e-05
46.1% identity (52.1% similar) in 1040 nt overlap (1-883:2-849)

      10      20      30      40
Rosal1 GGG---GCG---CCATCGTTCTCGGACCTTAT---CTGTTAAGTGAAGGAATGGGGTG
      ::  ::  ::  . ::  . ::  ::  ::  ::  ::  ::  ::  ::  ::
Rosal1 GTGTAAGCAGATACAAATTTTAGAGGAGCGTAAACCTCTTATATGA-----TG
      10      20      30      40      50

      50      60      70      80      90     100
Rosal1 TAAAGCAGATACAAATTTTAGAGGAGCGTAAACACGGGTCTTCGGAAGTAGATAGTTC
      ::::::::::::::::::::::::::::::::::::::::::::
Rosal1 TAAAGCAGATACAAATTTTAGAGGAGCGTAAAA-----TC
      60      70      80

      110     120     130     140     150     160
Rosal1 ATGCGGCTTAGGCAGGCACACGCTACAACCAAGTGACACCTCACGATCACTTCTCTGCGAG
      ::::: . :  ::::: . :  ::  ::  ::::  :  ::  . :  ::
Rosal1 ATGCGTTT---GCATGTA-----AAGCAGATACA-----AATT---TTAGAG
      90      100      110      120

      170     180     190     200     210     220
Rosal1 GAGCGCCAAGCTCAAGCGCCATCTAATGTATGTAAAGCAGATACAAATTTTAGAGGAGCG
      ::::: . :  ::::: . :  . :  ::::::::::::::::::::::::::::
Rosal1 GAGCGTAAAA--CAAGGT--ATC----AGATGTAAAGCAGATACAAATTTTAGAGGAGCG
      130      140      150      160      170

      230     240     250      260     270     280
Rosal1 TAAAAGCTGAAGTCGACCACG-----TTGTTCCCGCCCTAACCGGGGACACACTATA
      ::::: . :  ::::: . :  :  ::  ::  :  ::::: . :  ::
Rosal1 TAAAACGCTAACCCGACCACGCCCAATAGTAGTCCGCGCGGT----GTGGGCGG-CT-TT
      180      190      200      210      220

      290     300     310     320     330     340
Rosal1 TAATATCCAAGACAACCTAA-AGCTGCAAAATATTTAAAGCGAGAGGGTCCCGACCGTTTCG
      : :  ::  ::::: . :  ::  ::  ::  ::  ::  ::  ::
Rosal1 TGACCTC-----ACCTAACATCTG-----CGCCTTTTCT
      230      240      250

      350     360     370     380     390
Rosal1 ATTACACTTAGCACGAAACTT-GGTCGCT---TTCTTACCAT----CATACGGGGGAATA
```



```
1  # Name: Christopher A. Lee
2  # Date: 02/18/2025
3  # Prof: Dr. Callura
4  # Course: Bioinformatics – ENBC 455
5
6  from Bio import SeqIO
7  import Levenshtein
8  # Problem 3
9  with open('hw4_3.txt', 'r') as f:
10     seq1 = f.readline().strip()
11     seq2 = f.readline().strip()
12
13     count = 0
14     for i in range(len(seq1)):
15         if seq1[i] != seq2[i]:
16             count += 1
17
18     print("Problem 3\nHamming Distance:", count, "\n")
19
20 # Problem 4
21 with open('hw4_4.txt', 'r') as f:
22     seq = [str(fasta_seq.seq) for fasta_seq in SeqIO.parse(f, "fasta")]
23
24     dist = Levenshtein.distance(seq[0], seq[1])
25     print("Problem 4\nEdit Distance:", dist)
```

Problem 3

Hamming Distance: 496

Problem 4

Edit Distance: 304